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A hybrid approach for vision-based structural displacement measurement using transforming model prediction and KLT

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ABSTRACT

As civil infrastructures age, monitoring their health conditions has become increasingly critical. Dynamic displacement measurement is a prevalent method for assessing structural health. Traditional techniques, which often involve installing instruments and scaffolding, can interfere with the response of the structures. To address the challenge, non-contact measurement methods have been developed; however, these are typically costly and require expert operation. Advances in high-speed industrial cameras and image processing technology now enable vision-based displacement measurement. Despite their effectiveness, existing vision-based methods face significant limitations, including their inability to maintain tracking when line-of-sight is obstructed, sensitivity to lighting variations, and the need for manual intervention when feature points are lost. This study introduces a novel hybrid approach, termed ToMP-KLT, which combines the KLT tracker with a deep learning-based model. This method harnesses the precision of the KLT tracker under favorable conditions and the robustness of the deep learning-based tracker under adverse conditions. Its effectiveness is validated through simulation-based tests, lab-scale experiments, and field testing on Cheonsa Bridge, demonstrating substantial improvements in tracking robustness against occlusions and varying light conditions.

1. Introduction

Due to the deterioration of civil infrastructure, monitoring the condition of structures is becoming increasingly important. Recently, the First Church of Christ, built in 1853 in New London, collapsed. In 2023, the majority of the Davenport Hotel was suddenly demolished, causing millions of dollars in damage. In 2022, the 19th-century Morbi Bridge collapsed, resulting in more than 180 injuries. These events highlight the importance of assessing the health condition of structures. Structural displacement measurement is crucial for evaluating the health condition of infrastructures. Some characteristics of the structures, such as unit influence surface [1], deflection [2,3], natural frequency and mode shape [4,5] can be derived from displacement data. These characteristics facilitate early damage detection, helping to prevent negative outcomes [6,7]. Currently, structural displacement measurement continues to attract significant attention from researchers [8–17].

Various studies have been developed to measure the displacement of the structure. The linear variable differential transducer (LVDT) has traditionally been used to directly measure structural displacement. However, installing an LVDT necessitates additional tasks such as erecting a scaffold on tested structures which may affect the response of the structures as well as be expensive for

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large structures. In addition, its output includes some interference due to electromagnetic interference and stray capacitance [18]. Accelerometers are often used to measure the response of structures. The displacement can be calculated by double integrating acceleration [19,20]. However, this approach has limitations due to measurement noise and unknown initial conditions, leading to a low-frequency drift in the estimated displacement. Strain gauges are also used to measure strain or deformation in structures. This method relies on the relationship between displacement and strain mode shapes [21]. However, small noise can lead to significant relative errors in this technique if the amplitude level of the beam during off-resonant excitation is not sufficiently high. A displacement estimate technique combining accelerometer and strain gauge readings was recently proposed by Ma et al. [22]. While this performance has been verified, it is limited to structures that resemble beams.

On the other hand, structural displacement estimation can be accomplished using non-contact sensors such as laser Doppler vibrometer (LDV) [23,24], ultrasonic ranging system (URS) [25,26], global positioning system (GPS) [27,28], and light detection and ranging (LiDAR) [29,30]. The LVD can measure displacement with high accuracy, making it suitable for monitoring small changes in structures. However, LDV is too sensitive to the environment and not applicable for long-term monitoring. The URS is another type of non-contact method. It can measure over varying distances and surface characteristics, but the accuracy can be influenced by temperature and humidity. In addition, the GPS can provide accurate displacement. However, the GPS signals can be obstructed, leading to decreased accuracy. Besides, LiDAR can be used in structural displacement monitoring. This method provides high accurate measurement, but it requires significant processing power and expertise for data interpretation. Additionally, non-contact sensors tend to be not cost-efficient and typically require a team of experts to operate.

Recently, computer vision techniques are providing opportunities for measuring structural displacement by taking advantages of relatively low cost with capability of measuring multi point displacements. Unlike traditional methods, these techniques do not require expensive sensor setups, making them more cost-effective and easier to maintain. Due to their advantages, such as long-distance capabilities and ease of operation, various vision-based methods have been developed for many application scenarios. A visual tracking system [31] was proposed to monitor the vibration displacement at the center of the 1410 m span of the Humber bridge in the United Kingdom. Wadhwa et al. [32] proposed a technique for manipulating small movements in videos by analyzing motion through complex-valued image pyramids. This approach processed videos in the phase domain, supporting larger magnification factors with significantly less sensitivity to noise. The technique improved performance over traditional methods in various applications, such as scientific analysis and video enhancement, by offering higher motion magnification with fewer artifacts and better noise handling. Feng et al. [33] utilized a template matching technique based on orientation-code matching (OCM) to measure the dynamic response of civil structures. Guo et al. [34] developed a high-speed camera system for real-time displacement measurement. A method based on the Lucas–Kanade template matching algorithm was verified by the measurement results of a viaduct. Yoon et al. [5] employed consumer-grade cameras and the Kanade–Lucas–Tomasi (KLT) feature tracker [35,36] to identify structural displacement during shake table tests. This method involves selecting a region of interest (ROI) and tracking all feature points within that region. Zhao et al. [37] combined KLT and support correlation filters (SCF) to improve the robustness for tracking under earthquake ground motions. Jana et al. [38] introduced a method for estimating cable tension in the Dubrovnik cable-stayed bridge using video recorded by a handheld camera, addressing the challenges of camera movement with image processing techniques. This method used the KLT feature tracking and random sample consensus (RANSAC) algorithm to stabilize the video, followed by phase-based motion estimation and short-time Fourier transform (STFT) to determine real-time frequency variations and cable tension. The method was tested on the Franjo Tudjman Bridge and successfully demonstrated accurate cable tension estimation, proving its potential for effective structural health monitoring. Lee et al. [39] proposed a method for measuring structural displacements using a camera mounted on the structure itself, as opposed to other methods where the camera is installed outside the structure to track feature points. The recent advancements in vision-based measurement have provided a valuable opportunity to accurately measure the dynamic movement of structures. This approach offers the advantage of being more cost-effective and accessible compared to conventional methods.

Although prior studies have demonstrated the utility of these techniques in measuring structural displacement, several limitations have been identified. Firstly, conventional vision-based methods cannot track an object when an obstacle interrupts the line of sight between the camera and the object. Due to their reliance on feature points, these methods fail to maintain tracking when unexpected obstacles result in the loss of these points. Secondly, these methods are sensitive to lighting variations. Particularly under low-light conditions, such as those prevalent in the evenings, their accuracy significantly diminishes. Lastly, when feature points are lost, conventional methods necessitate a manual intervention to redefine the region of interest, as the tracking algorithms lack the capability to automatically adjust the region of interest once tracking is disrupted.

To address some of the limitations mentioned above, several researchers have explored the use of convolutional neural networks (CNN) for measuring structural displacement, capitalizing on their robustness in various computer vision tasks. Dong et al. [40] proposed a novel structural displacement measurement method using deep learning-based full field optical flow methods. Statistical analysis of the comparative results showed that this method gives higher accuracy than the traditional optical flow algorithm and shows consistent results in compliance with displacement sensor measurements. Huang et al. [41] proposed a vision-based method called DAVIM, which is enhanced by deep learning for measuring structural displacements. This method incorporates deep learning algorithms such as CNN and generative adversarial network (GAN) to improve accuracy and robustness in harsh environments. DAVIM is validated through three experiments involving periodic vibration tests, wind tunnel tests, and field measurements. The results show that DAVIM outperforms traditional sensors and the vision-based vibration measurement method, especially in measuring large displacements with rigid-body rotational components and in long-term field measurements. Jeon et al. [42] proposed using CNNs to automatically track cables and redefine the region of interest autonomously. Similarly, Pan et al. [43] have developed CNN-based displacement measurement methods, employing networks such as YOLOv3-tiny and YOLOv3-tiny-KLT,

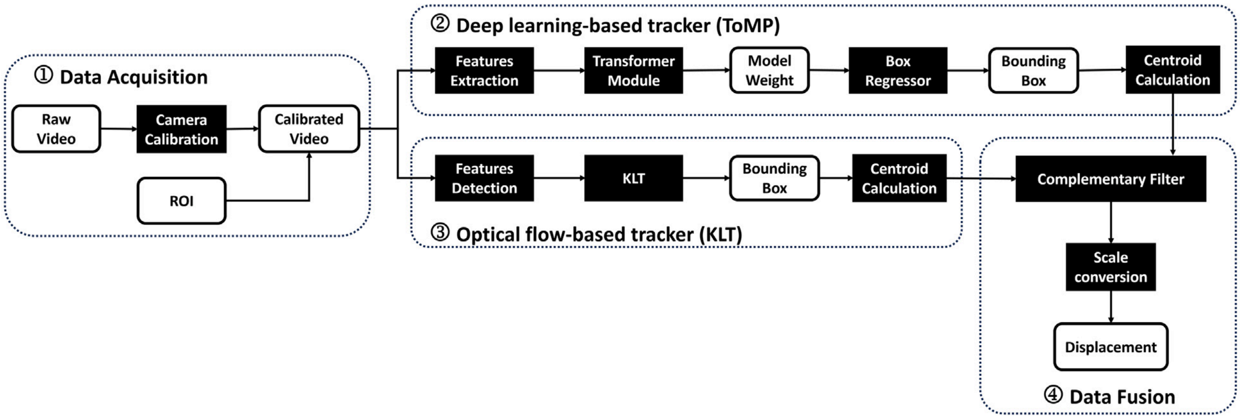

Fig. 1. Overview of the proposed ToMP-KLT system.

to measure structural motion. Lin et al. [44] introduced a novel method featuring a high-precision displacement measurement algorithm that utilizes multiscale feature extraction and fusion. Additionally, Siamese networks, including the Siamese region proposal network (SiamRPN) [45,46] and SiamMask [47], have been employed for structural displacement measurement. These methods showed effectiveness in measuring lower frequency response. However, most of these studies were confined to specific structural components, such as cables, and were unable to cope with extreme environmental conditions. Furthermore, the accuracy of the CNN-based structural displacement measurement was lower than that of conventional method under favorable environmental conditions, especially for the high frequency response.

This study aims to develop an innovative vision-based structural displacement measurement method named as ToMP-KLT by integrating the conventional KLT tracker with a deep learning-based model such as transforming model prediction (ToMP) [48]. This proposed method leverages the precision of the KLT tracker under favorable environmental conditions and the effectiveness of Transformer-based approaches under extreme conditions. Transformer-based can incorporate a self-attention mechanism to capture contextual information, thus can enhance robustness and accuracy in challenging scenarios, such as when the line of sight is obstructed, or lighting conditions are suboptimal. Additionally, the proposed method features automatic redefinition of the region of interest, allowing it to reinitiate tracking of an object following a tracking failure. The proposed method consists of four main steps. Initially, to eliminate radial distortion, data is obtained through a camera calibration procedure. Subsequently, a Transformer-based model tracks the target and provides the tracking results. The displacements are then estimated using the KLT algorithm. Finally, the displacement values are calculated by fusing these two measurements using a complementary filter. The effectiveness of this method is demonstrated through simulation-based experiments, lab-scale experiments, and on-site testing at Cheonsa Bridge.

The contributions of this paper can be summarized as follows:

(1) A novel approach for structural displacement measurement combining the conventional tracker (KLT) together with a recent deep learning-based model (ToMP) was proposed.

(2) The proposed method improves the overall tracking robustness against occlusion and brightness conditions.

(3) The proposed method can automatically redefine the region of interest after the failure.

(4) The effectiveness of the proposed method is demonstrated through simulation-based experiments, lab-scale experiments, and an on-site test.

The remaining sections of this paper are organized as follows: Section 2 outlines the proposed method in this study. Section 3 discusses the experiments and results, including the experimental setup, environment, data acquisition, and comparison algorithm. Finally, Section 4 provides the conclusion of this study.

2. Development of the proposed system

The proposed methodology for measuring structural displacement named ToMP-KLT consists of four steps, as shown in Fig. 1. First, data is acquired through a camera calibration process to remove radial distortion. Subsequently, a Transformer-based model is used to track the target and provide the tracking result. Next, another tracking result is estimated by KLT algorithm. Finally, the displacement can be calculated by fusing these two values with complementary filter. The following sections will provide further details on each step.

2.1. Data acquisition

Camera calibration is the initial stage in measuring displacement. The goal of this phase is to remove any radial distortion from the images, which is a common problem with wide-angle lens cameras. Radial distortion was eliminated using Zhang [49] flexible

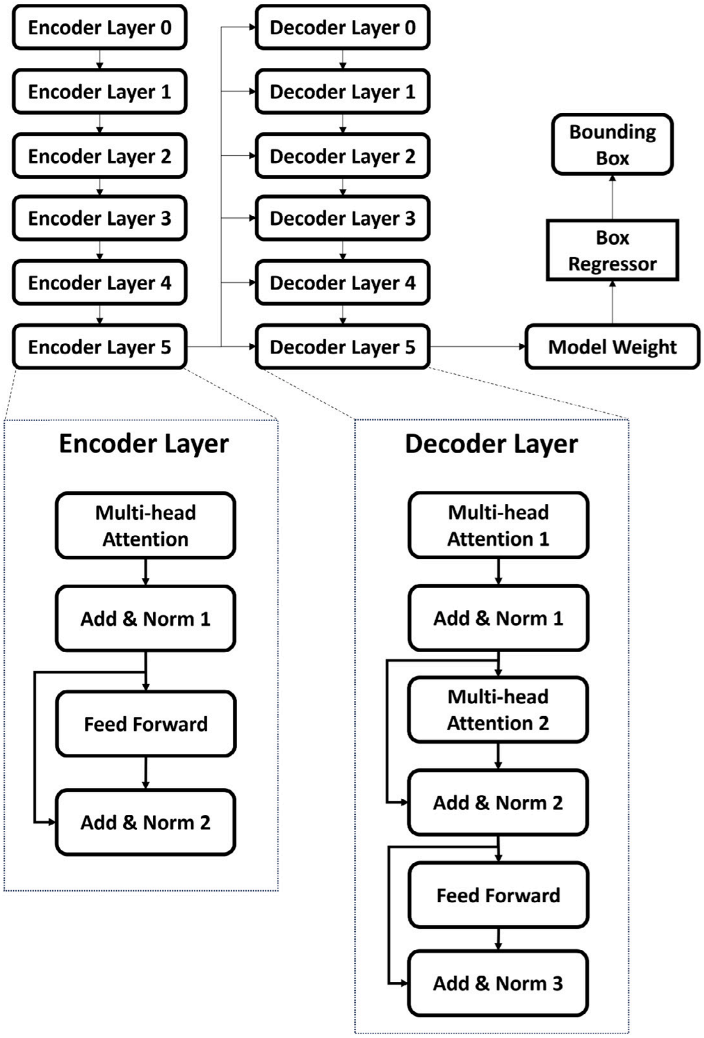

Fig. 2. Architecture of the Transformer-based model.

camera calibration technique. The focal length (f), radial distortion (k_1 and k_2), rotation matrix (R), and translation vector (T) are all accurately predicted by this method as the following equations:

$$P = RX + T \tag{1}$$

where X belongs to the global coordinates and P is the projection with homogeneous coordinates. The perspective is calculated as follows:

$$p = -P/P_z \tag{2}$$

Next, the radial distortion is determined by $r(p)$

$$r(p) = 1.0 + k_1\|p\|^2 + k_2\|p\|^4 \tag{3}$$

Finally, the undistorted coordinates p' is considered as follows:

$$p' = f \cdot r(p) \cdot p \tag{4}$$

2.2. Deep learning-based tracker

The second step is the deep learning-based tracker shown in Fig. 2. In this architecture, the features are first extracted by the ResNet-50 backbone [50]. The transformer encoder then generates improved features by reasoning globally across frames,

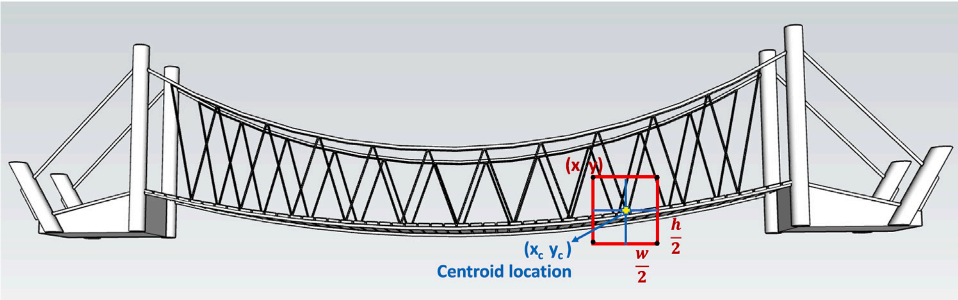

Fig. 3. The illustration of Centroid calculation.

which enhances the robustness of the proposed approach. The transformer encoder represents the key components of each image frame using multi-headed self-attention modules. Furthermore, the model weight is generated by the transformer decoder using the improved features. Finally, the model is applied on the enhanced features to predict the bounding box coordinates.

ToMP was trained on large-scale datasets with data augmentation techniques. COCO [51] and ImageNet [52] are distinguished by the diversity and large quantity of static images, which allows models to learn numerous features from complex contexts. However, the models lack the ability to track objects continuously across frames. In contrast, GOT-10k [53], and LaSOT [54] consist of videos, allowing models to learn how to track objects over extended periods continuously. GOT-10k and LaSOT are particularly strong in long-term object tracking. TrackingNet [55] provides high realism from real-world videos with various objects and contexts, but the inconsistent video quality can present challenges for tracking. However, the datasets listed above only contain positive pairs, which may generate an imbalance in the dataset distribution and lead to drifting to the erroneous object because the training dataset only distinguishes between the object and the background. To overcome this issue, negative pairs were added into the training dataset for the same category. This method prevented the tracker from wandering to the wrong item when it needed to re-detect it after occlusion and being out of view. The computation time is accelerated on an Ubuntu 22.04 system with the NVIDIA GeForce GTX 3060. The learning rate of 0.0001 was used with 300 epochs. It takes about 0.5 s to calculate the displacement in each frame. The displacement of the object can be known as the change of centroid location in the bounding box compared to the initial frame. From the output of the Transformer-based model, the centroid location as in Fig. 3 is easily achieved by the following equation.

$$x_c = x + \frac{w}{2} \tag{5}$$

$$y_c = y + \frac{h}{2} \tag{6}$$

2.3. Optical flow-based tracker

The third step in the proposed system is the optical flow-based tracker. To find the characteristics of the target, Harris corner detection was used to determine the presence of a corner while moving the window to obtain the feature points [56]. The sum of squared differences E for window W is then obtained as follows:

$$E_{x,y} = \sum_{u,v} W_{u,v} \left[I_{x+u,y+v} - I_{u,v}\right]^2 = \begin{bmatrix} x & y \end{bmatrix} \mathbf{M} \begin{bmatrix} x \\ y \end{bmatrix} \tag{7}$$

where $I_{u,v}$ and $I_{x+u,y+v}$ are the image intensities of the initial video frame and the shifted frame, respectively. Symmetric matrix $\mathbf{M}$ is the section moment that obtained as

$$\mathbf{M} = \begin{bmatrix} \left(\frac{\partial I}{\partial x}\right)^2 & (\frac{\partial I}{\partial x})(\frac{\partial I}{\partial y}) \\ (\frac{\partial I}{\partial x})(\frac{\partial I}{\partial y}) & \left(\frac{\partial I}{\partial y}\right)^2 \end{bmatrix} \tag{8}$$

Consider R as the measurement of corner response. Utilizing $det(M)$ and $Tr(M)$ in the formulation is advantageous as it eliminates the requirement for an explicit eigenvalue decomposition.

$$R = \det(M) - k \cdot \mathrm{Tr}(M)^2 \tag{9}$$

here, k is a parameter. R will have high positive values in the corner region, negative in the edge regions, and small in the flat region.

After selecting the features in the initial frame, we utilize the KLT algorithm to track these point features throughout the entire video sequence. The image intensity of the current frame, denoted as $J(x)$, is represented using the intensity of the previous frame, denoted as $I(x)$, according to Eq. (10), assuming a small motion for the features.

$$J(x) = I(x - d) = I(x) - g \cdot d \tag{10}$$

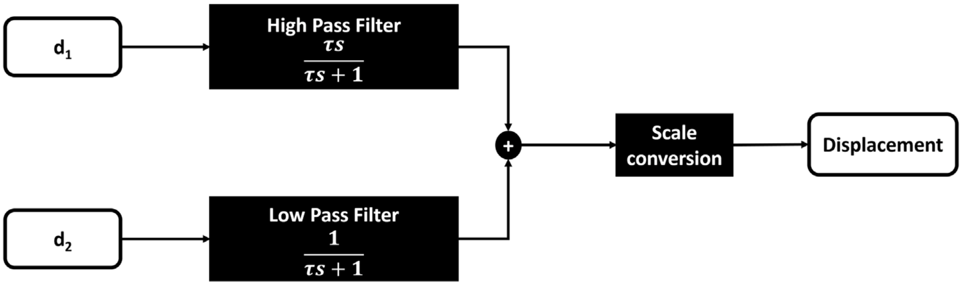

Fig. 4. Overview of the complementary filter in ToMP-KLT.

where d represents the displacement vector between two consecutive video frames, while the gradient vector $g = (\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y})$. The residue ϵ for a small window of pixels surrounding the feature can be defined by the following equation:

$$\epsilon = \int [I(x-d) - J(x)]^2 \, wdA = \int [I(x) - g \cdot d - J(x)]^2 \, wdA = \int (h - g \cdot d)^2 \, wdA \tag{11}$$

In this equation, w represents a weighting function, and $h = I(x) - J(x)$. To minimize the residue, the previous equation can be differentiated with respect to d and then set the result equal to zero as follows:

$$\frac{d\epsilon}{dd} = \int (h - g \cdot d) gwdA = 0 \tag{12}$$

Finally, the displacement vector d can be calculated by the following equation.

$$d = G^{-1} e \tag{13}$$

where $G = \int gg^{\mathrm{T}} \, wdA$ and $e = \int (I - J) gwdA$.

2.4. Data fusion based on complementary filter

The effectiveness of the KLT algorithm in tracking high-frequency displacements stems from its ability to precisely track small, rapid changes in feature point positions across consecutive video frames. This high-frequency resolution allows the KLT algorithm to capture fine-grained movements with considerable accuracy. On the other hand, deep learning models, particularly those based on transformer architectures, excel in tracking low-frequency displacements due to their capability to capture long-term dependencies and contextual information across multiple frames. The self-attention mechanism in transformers enables them to integrate information over longer sequences, thus maintaining robustness against occlusions and variations in lighting conditions. To obtain the best measurement, complementary filter is required as a fusion. In Fig. 4, d_1 and d_2 are the displacement vector estimated from KLT and ToMP, respectively. The result d produced by the filter as follow:

$$\mathbf{d} = \frac{\tau s}{\tau s + 1} d_1 + \frac{1}{\tau s + 1} d_2 \tag{14}$$

where τ is the time constant. The selection of the time constant is performed through empirical experiments and system performance observations to ensure optimal accuracy and stability. The complementary filter is a pre-defined static filter, with $\tau = 3$ s and remaining unchanged during operation.

$$scalefactor = \frac{D}{\mathbf{d}} \tag{15}$$

Finally, the displacement can be obtained by scale conversion. The scale factor in this step is the relationship between physical distance (D) and the pixel distance ($\mathbf{d}$).

3. Experiments and results

To validate the effectiveness of the proposed system, three types of experiments were conducted. Firstly, a simulation-based test was used to determine the accuracy of the proposed method under challenging conditions. Next, a laboratory experiment was conducted to confirm the effectiveness of the proposed method on a lab-scale test. Finally, an on-site experiment was carried out to demonstrate the practicality of the approach.

3.1. Simulation-based test

A simulation-based test was conducted to validate the proposed method under challenging conditions. The response of a 3-story building was generated using MATLAB code, with a checkerboard placed on the second floor as a target for object trackers. The resulting animation was then converted into a 560 × 420 video with a frame rate of 40 fps.

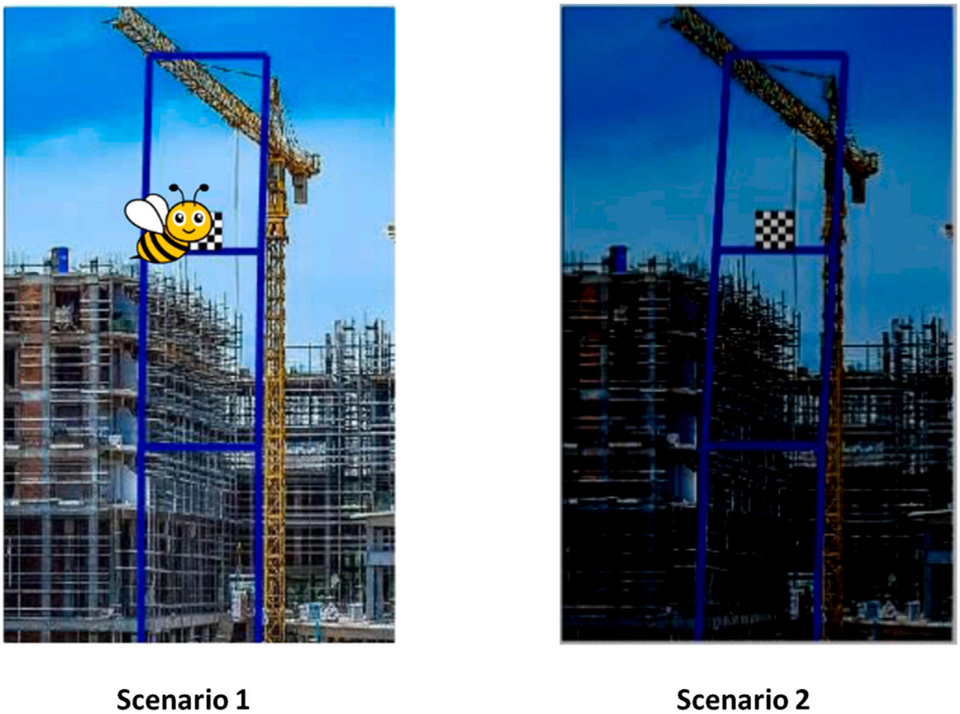

Fig. 5. Two scenarios for the simulation-based validation test. Scenario 1 with obstacle and scenario 2 with poor lighting condition.

Table 1
Comparison between the proposed approach and conventional methods for the simulation-based validation test.

Method	RMSE (pixel)		NRMSE (%)	
	Scenario 1	Scenario 2	Scenario 1	Scenario 2
KLT	268.584	11.443	1.431	0.093
SiamRPN	90.179	20.659	0.047	0.167
FlowNet2	44.767	18.657	0.239	0.152
ToMP	44.441	17.511	0.236	0.142
Proposed method (ToMP-KLT)	8.889	3.841	0.047	0.031

Monitoring civil structures presents unique challenges, particularly when obstructive objects obscure the view and during low-light conditions. These challenges can significantly hinder the effectiveness of monitoring systems, especially those relying on cameras. Obstacles such as construction materials, animals, and vegetation can obstruct the view of cameras. Traditional cameras may struggle to produce usable footage in low-light environments, leading to decreased monitoring effectiveness and potentially missing critical events or structural changes. To account for these challenges, two scenarios were added to the simulation, as depicted in Fig. 5.

Error analysis was carried out using the root mean square error (RMSE) and normalized root mean square error (NRMSE) as follows to compare the measurement effects of the proposed method:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{k}(y_i - \hat{y}_i)^2} \tag{16}$$

$$NRMSE = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{k}(y_i - \hat{y}_i)^2}}{y_{max} - y_{min}}, \tag{17}$$

where y_i is the reference value at the ith second; $\hat{y}_i$ is the measurement result at the ith second; n is the ordinal number of a video frame. Table 1 presents the RMSE and NRMSE values of the measurements. The results indicate a significant performance improvement in the proposed method compared to the conventional KLT tracker algorithm.

Fig. 6 shows the tracking results for scenario 1, where a bee with a free-flight trajectory was added to the frame. The proposed method demonstrates the highest accuracy in estimated displacement, with the proposed method nearly perfectly aligning with the ground truth. In contrast, other methods such as KLT, SiamRPN, FlowNet2, and ToMP show significant deviations from the ground truth. Specifically, KLT exhibits large discrepancies, particularly after 2.5 s, with a maximum deviation of around 600

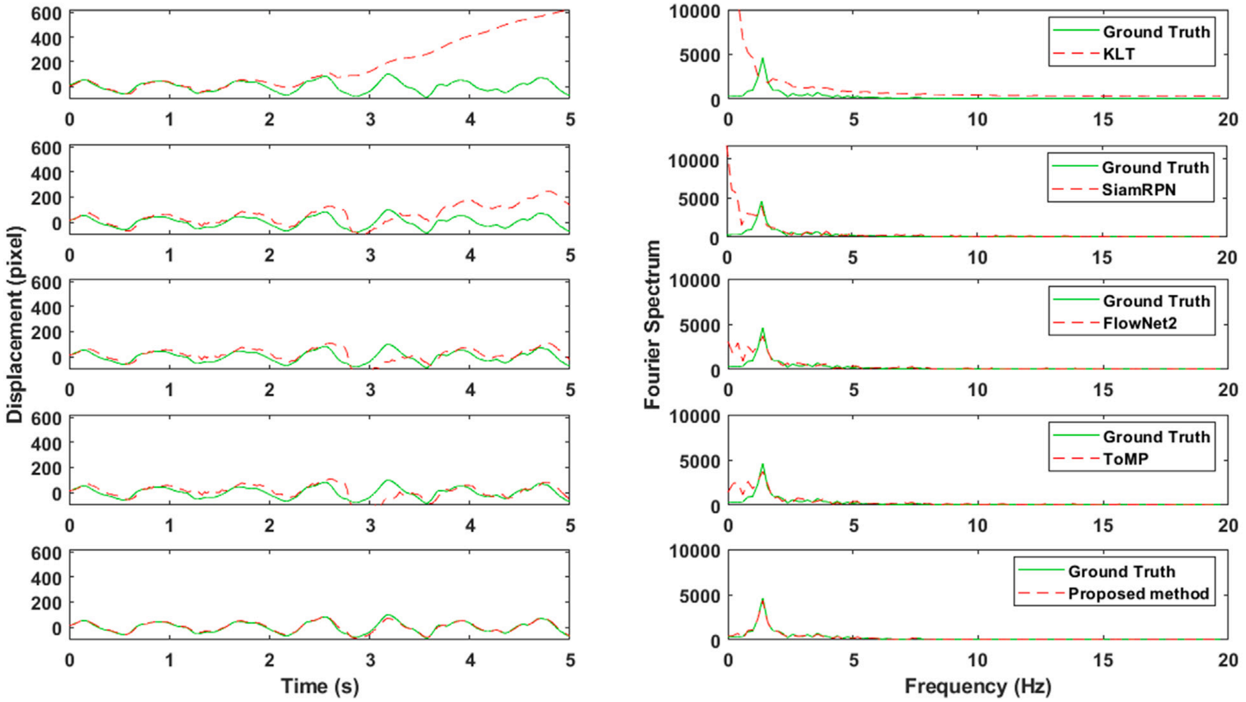

Fig. 6. Estimated displacement and Fourier spectrum for the scenario 1.

pixels. SiamRPN, FlowNet2, and ToMP also display considerable errors, with peak deviations of approximately 100 pixels. In the frequency domain analysis, the Fourier spectrum of the proposed method closely matches the ground truth across all frequency ranges. Conversely, the spectra for these other methods exhibit mismatches, particularly at higher frequencies, with extra peaks not corresponding to the ground truth. The proposed method consistently achieves the lowest RMSE and NRMSE values, underscoring its superior accuracy and precision. For Scenario 1, the proposed method records an RMSE of 8.889 pixels and an NRMSE of 0.047, compared to KLT 268.584 pixels RMSE and 1.431 NRMSE.

Fig. 7 shows the scenario of fluctuating brightness levels in the environment. In this case, the oscillation pattern remains the same as before, but the light gradually decreases to zero and then increases again. In this scenario, other methods such as KLT, SiamRPN, FlowNet2, and ToMP still show discrepancies, though less pronounced than in Scenario 1. Specifically, KLT exhibits deviations that reach up to 50 pixels, indicating significant errors. SiamRPN, FlowNet2, and ToMP also display inaccuracies, with deviations around 50 pixels at their peaks. The frequency domain analysis reveals that the proposed method closely matches the ground truth across all frequency ranges, similarly to Scenario 1. The spectra for KLT, SiamRPN, FlowNet2, and ToMP, although improved, still exhibit mismatches, particularly at higher frequencies, with extra peaks not corresponding to the ground truth. The proposed method maintains its superior performance with an RMSE of 3.841 pixels and an NRMSE of 0.031, while KLT shows 11.443 pixels RMSE and 0.093 NRMSE. This highlights its superior performance in low-light conditions. The proposed method can automatically redefine the region of interest after the failure.

3.2. Lab-scale test

Two lab-scale tests were conducted to validate the performance of the proposed method in controlled environments: test 1 with a 3-story building model, and test 2 with a bridge model.

For the lab-scale test 1, the target was mounted on the 3-story building as in Fig. 8. The reference displacement was measured by a highly sensitive displacement transducer (CDP-50) with a National Instruments NI-9230 module. The information about the devices can be found in Table 2. The camera was placed on a tripod. The distance from the camera to the structure is 1.0 m. The vibration was made by using a hammer.

The analysis of Fig. 9 and Table 3 provides a comprehensive evaluation of various displacement measurement methods for a 3-story building model and a bridge in a lab-scale test, highlighting the superior performance of the proposed method. In Fig. 9, the time domain analysis demonstrates that the proposed method closely aligns with the ground truth, with minimal deviations. Specifically, the KLT method exhibits significant discrepancies, particularly after 4.5 s, with deviations reaching up to 60 mm. SiamRPN and FlowNet2 methods also show notable errors, with deviations around 10 mm at their peaks. ToMP performs better but still presents minor deviations within 5 mm. The frequency domain analysis further corroborates these findings, showing that the proposed method closely matches the ground truth across all frequency ranges. In contrast, the spectra for KLT, SiamRPN,

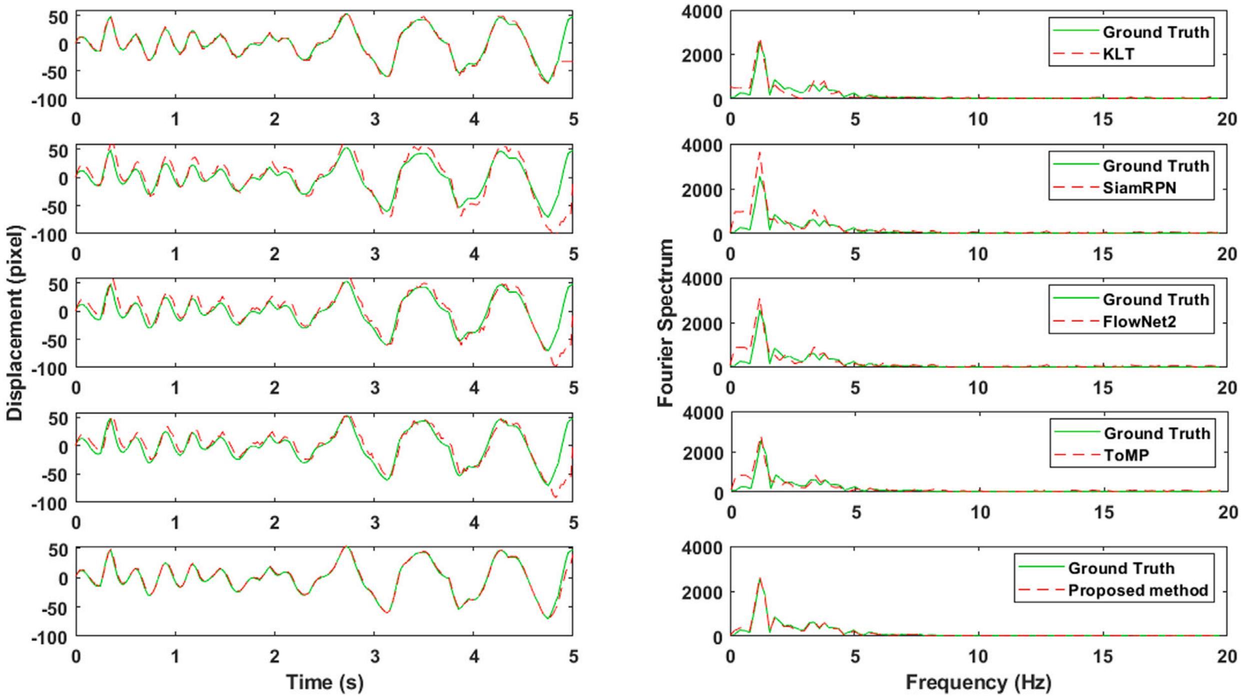

Fig. 7. Estimated displacement and Fourier spectrum for the scenario 2.

Table 2
Descriptions for the devices used in the lab-scale tests.

Type	Description
Camera	Model: FLIR BFS-PGE-23S3C Resolution: 1920 × 1200 Frame rate: 59 Interface: USB3
Displacement transducer	Model: CDP-50 Capacity: 50 mm Sampling rate: 1024

FlowNet2, and ToMP exhibit mismatches, particularly at higher frequencies, with additional peaks not corresponding to the ground truth. Table 3 quantitatively reinforces these observations by presenting the RMSE and NRMSE for different methods. For the 3-story building model, the proposed method achieves the lowest RMSE of 0.318 mm and an NRMSE of 0.025, compared to KLT, RMSE of 22.166 mm and NRMSE of 1.759. Other methods like SiamRPN, FlowNet2, and ToMP, though better than KLT, still lag significantly behind the proposed method.

In the lab-scale test 2, a bridge model was set up at the Chungbuk National University. The configuration of the test is shown in Fig. 10. The reference displacement was measured using CDP-50, which was mounted under the bridge. The camera was mounted on a tripod 3.3 m from the structure. In this case, the ROI was not a checkerboard. It is important to note that the proposed method does not require any additional training steps, unlike most tracking methods which require transfer learning with a new dataset. The results of the tracking can be seen in Fig. 11. In the time domain analysis, the proposed method closely aligns with the ground truth, indicating high accuracy in estimating displacement over time. In contrast, the KLT method exhibits significant discrepancies, particularly towards the end of the observation period, with deviations reaching up to 2 mm. SiamRPN also displays notable errors, with consistent deviations throughout the duration. While the FlowNet2 and ToMP method perform better, the results still shows minor discrepancies. The proposed method, however, demonstrates minimal deviations, ensuring the most accurate displacement measurements. The ToMP-KLT maintains its superior performance with an RMSE of 0.217 mm and an NRMSE of 0.039, whereas KLT shows an RMSE of 1.211 mm and an NRMSE of 0.217. SiamRPN, FlowNet2, and ToMP, while improved over KLT, still do not match the precision of the proposed method.

3.3. On-site test

To validate the practicality of the proposed method, an on-site test was conducted at the Cheonsa Bridge in South Korea, as shown in Fig. 12. To measure characteristics of the cable, a machine vision camera was installed on the opposite side of the cable.

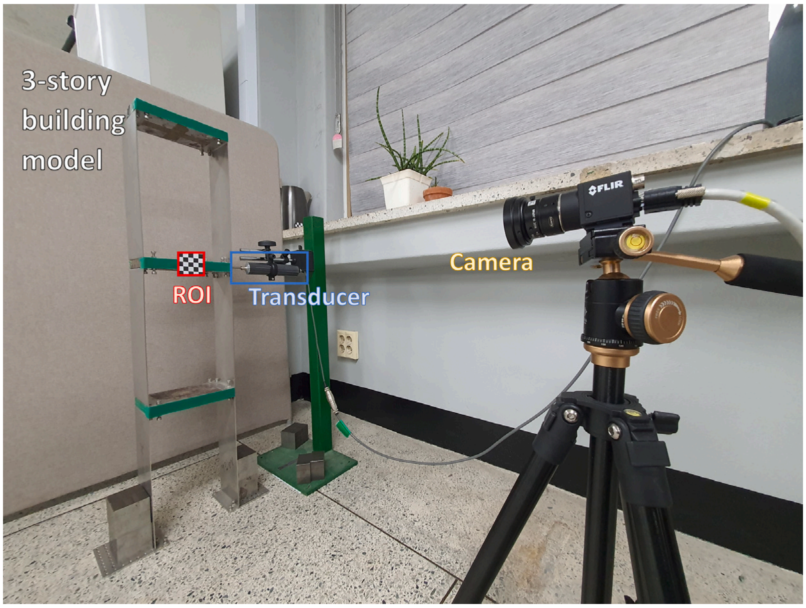

Fig. 8. Configuration of the lab-scale test 1 with a 3-story building model.

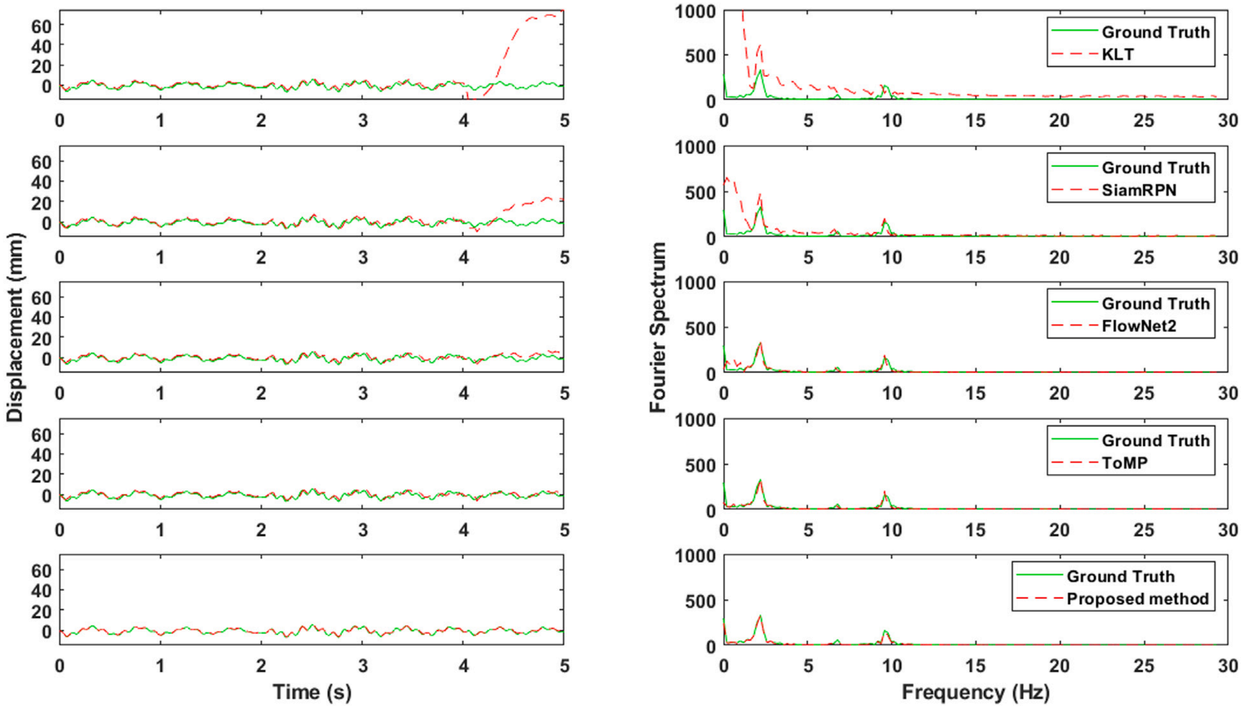

Fig. 9. Estimated displacement and Fourier spectrum for the lab-scale test 1 (3-story building model).

The distance from the camera to the cable is 15.0 m. The camera was set to record video with a resolution of 800 × 200 pixels at a frame rate of 59 frames per second. The specific section of the cable to be measured was positioned at the center of the camera view, ensuring accurate and consistent data collection.

Fig. 13 presents the results of an on-site validation test for estimated displacement, comparing multiple methods including KLT, SiamRPN, FlowNet2, ToMP, and the proposed method. Notably, a car passes during the 1st and 2nd second, creating a disturbance. At the 1st second mark, all methods still align reasonably well, but by the 2nd second mark, the KLT method shows a significant

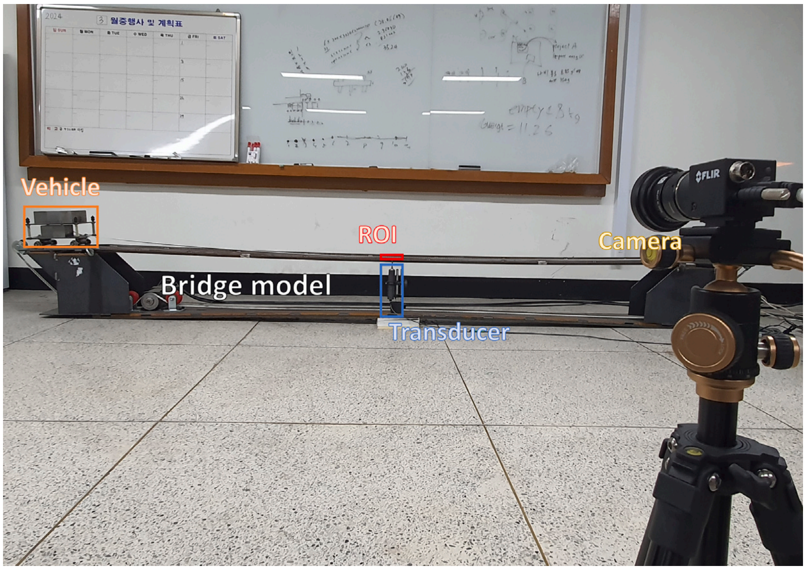

Fig. 10. Configuration of the lab-scale test 2 with a bridge model.

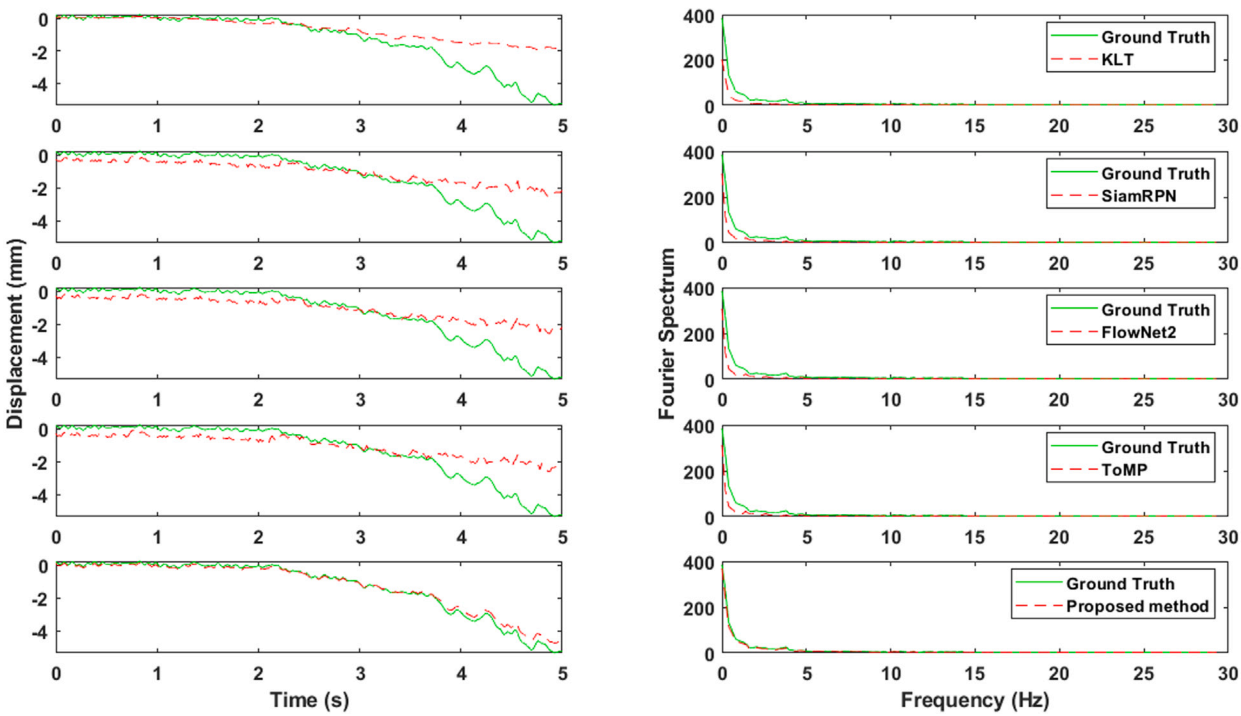

Fig. 11. Estimated displacement and Fourier spectrum for the lab-scale test 2 (bridge model).

deviation, with a noticeable drop in displacement. This is highlighted by the dotted yellow line representing KLT, which deviates sharply from the other methods. The proposed method maintains a close alignment throughout the duration, indicating superior performance in accurately tracking displacement. The Fourier spectrum chart illustrates the frequency content of the displacement signals. The proposed method closely matches all frequency ranges, similar to other methods like SiamRPN, FlowNet2, and ToMP, which also perform well. However, KLT shows some discrepancies in the higher frequency ranges, which could indicate less accuracy in capturing certain vibrational modes. The images at the bottom of Fig. 13 provide a visual validation of the displacement at different time intervals (0 s, 1 s, and 2 s). These images highlight the ROI used for tracking the displacement. The visual alignment of the ROI at different times supports the quantitative results shown in the charts. The ROI remains consistently aligned in the images

Table 3
Comparison between the proposed approach and conventional methods on the lab-scale test.

Method	RMSE (mm)		NRMSE (%)	
	3-story	Bridge	3-story	Bridge
KLT	22.166	1.211	1.759	0.217
SiamRPN	7.105	1.091	0.564	0.195
FlowNet2	2.114	1.087	0.168	0.195
ToMP	1.592	1.084	0.126	0.194
Proposed method (ToMP-KLT)	0.318	0.217	0.025	0.039

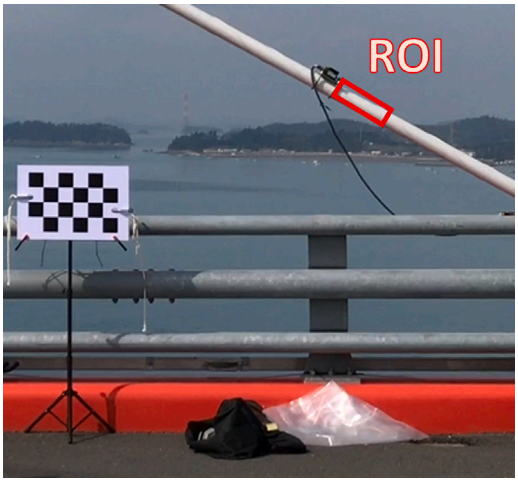

Fig. 12. Configuration of the on-site validation test.

corresponding to the proposed method, further validating its accuracy. In conclusion, the combined analysis of the time domain, frequency domain, and visual validation in Fig. 13 underscores the superiority of the proposed method. It consistently aligns both in terms of displacement over time and frequency content. The significant deviation observed in the KLT method, particularly at the 2nd second mark, contrasts sharply with the performance of the proposed method, which maintains accuracy throughout the test. This comprehensive evaluation demonstrates the reliability and effectiveness in accurately measuring displacement in on-site validation tests, making it a robust choice for real-world structural health monitoring applications.

4. Conclusion

This paper proposed a method for addressing the problem of existing vision-based displacement measurement. The proposed approach incorporates transformer-based model to automate the selection of ROI in real-world scenarios. Through a series of experiments, the superiority of the proposed method over conventional KLT is demonstrated. The results show that the proposed method can improve overall tracking robustness, even when facing occlusion or variations in brightness. Additionally, it is noteworthy that the proposed method maintains accurate tracking even when the surface of the structures has limited available features. The ToMP-KLT method is expected to make a significant contribution to the field of structural displacement measurement. The proposed method can improve tracking robustness and the ability to automatically redefine the region of interest after failure.

While the ToMP-KLT method has demonstrated promising results in enhancing the robustness and accuracy of vision-based structural displacement measurements, further research is necessary to expand its capabilities and address the limitations encountered during experiments. The authors will focus on extending the proposed method to acquire full-field displacement responses by developing techniques such as digital image correlation and other full-field displacement measurement methods [57–62]. This will enhance the ability to monitor and assess the health of large-scale structures comprehensively. Additionally, integrating the ToMP-KLT method with other sensing technologies, such as LiDAR, GPS, and accelerometers, could improve the accuracy and reliability of displacement measurements through a multi-sensor approach. Enhancing real-time processing capabilities and developing automated systems for real-time monitoring and data analysis will be crucial for practical field applications, enabling timely detection of structural issues and prompt maintenance interventions. Moreover, further research is needed to improve performance under extreme environmental conditions, such as heavy rain, fog, and extreme temperatures, by developing adaptive algorithms that can adjust to these conditions. Finally, implementing the ToMP-KLT method for long-term structural health monitoring requires effective data management strategies. By addressing these areas, the ToMP-KLT method can be refined and expanded, providing a more comprehensive and reliable solution for structural health monitoring and ensuring the safety and longevity of civil infrastructure.

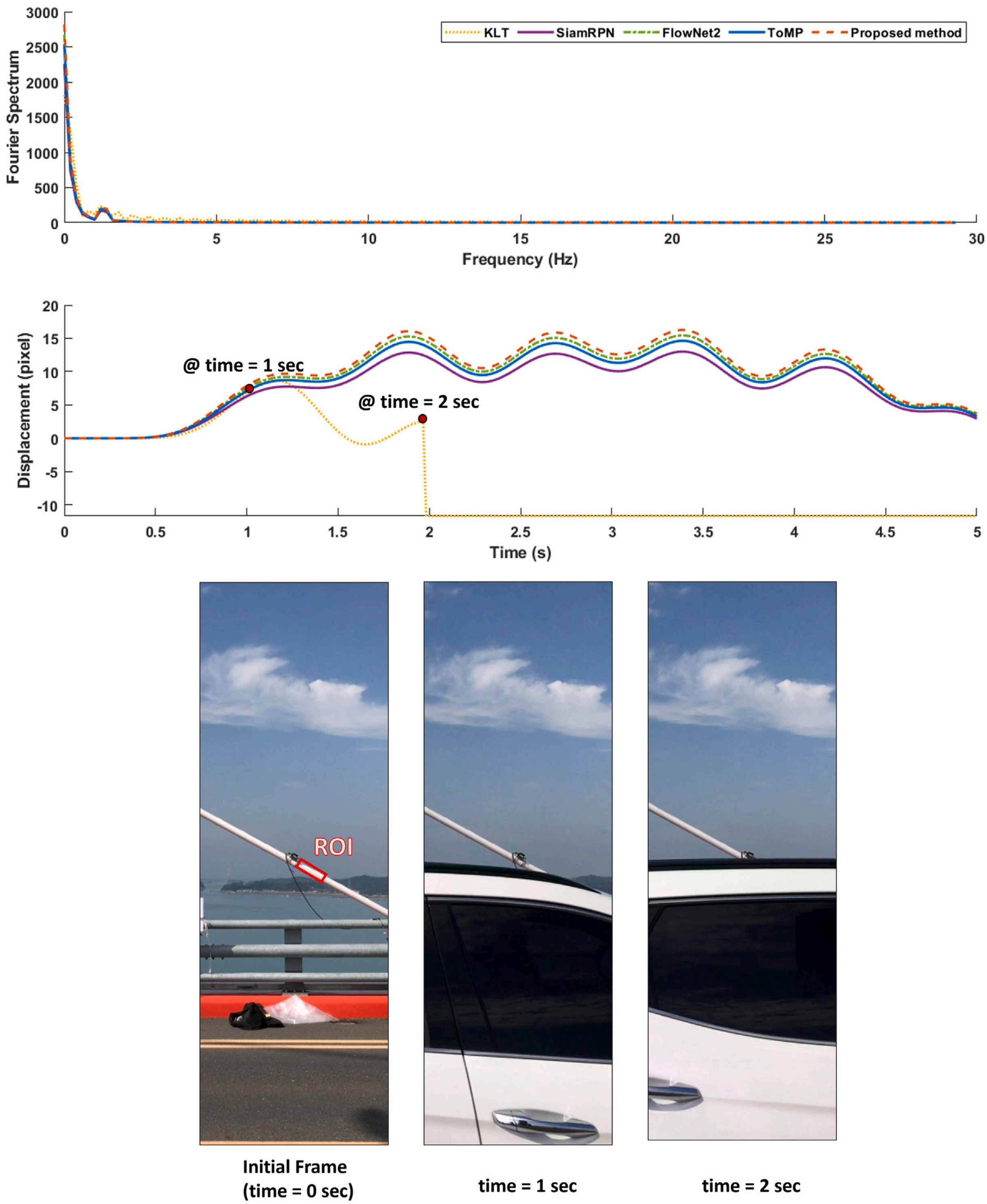

Fig. 13. Estimated displacement and Fourier spectrum for the on-site validation test.

CRediT authorship contribution statement

Xuan Tinh Nguyen: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Geonyeol Jeon:** Resources. **Van Vy:** Writing – review & editing. **Geonhee Lee:** Resources. **Phat Tai Lam:** Writing – review & editing. **Hyungchul Yoon:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hyungchul Yoon reports financial support was provided by National Research Foundation of Korea. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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